DS&AI

Machine Learning

Classification

DPP: 1

Q1 Find the man of 0-1 loss for the given predictions:

Y	F(X)
Cat	Cat
Cat	Dog
Dog	Panda
Panda	Panda
Rat	Dog
Rate	Rat

Round upto 1 decimal places

Q2 What does the ROC curve represent?
(A) The trade-off between precision and recall.
(B) The relationship between accuracy and Fl-score.
(C) The performance of a model across various thresholds.
(D) None of these

Q3 Imagine you have a dataset that contains information about a binary classification model. The model classifies instances into two classes: "Positive" and "Negative". After evaluating the model, you have the following results:
True Positives (TP): 120
False Positives (FP): 25
True Negatives (TN): 800
False Negatives (FN): 30
Calculate the precision of the model using the given information.
(A) 0.75 (B) 0.80
(C) 0.82 (D) 0.85

Q4 A machine learning model is trained to identify whether an email is spam or not. After evaluating the model on a test dataset, the following results are obtained
True Positive (TP): 175
True Negative (TN): 310
False Positive (FP): 123
False Negative (FN): 147
Calculate Precision?

Q5 How many statements are true?
(A) Type I error: The number of instances incorrectly predicted as positive
(B) Type II error: The number of instances correctly predicted as negative.
(C) Type I error: The number of instances incorrectly predicted as negative.
(D) Type II error: The number of instances incorrectly predicted as negative

Q6 You are designing a deep learning system to detect driver fatigue in cars. It is crucial that your model detects fatigue, to prevent any accidents. Which of the following is the most appropriate evaluation metric?
(A) Accuracy
(B) Precision
(C) Recall
(D) Loss vlaue

Q7 Which of the following are supervised learning problems? (multiple may be correct)
(A) Learning to drive using a reward signal
(B) Predicting disease from blood sample.
(C) Grouping students in the same class based on similar features.
(D) Face recognition to unlock your phone.

Q8 Which of the following are classification problems? (multiple may be correct)
(A) Predict the price of a house.
(B) Predict the disease of a patient.

(C) Predict the temperature for tomorrow.
(D) Has the student Learned to play a guitar yes/No

Q9 What is the use of Validation dataset in Machine Learning?
(A) To train the machine learning model.
(B) To evaluate the performance of the machine learning model
(C) To tune the hyperparameters of the machine learning model
(D) None of the above.

Q10 Given a training data set of 10,000 instances, with each input instance having 17 dimensions and each output instance having 2 dimensions, the dimensions of the design matrix used in applying linear regression to this data is
(A) 10000 * 17 (B) 10002 * 17
(C) 10000 * 18 (D) 10000 * 19

Q11 Logistic regression is robust to outliers. Why?
(A) The squashing of output values between [0, 1] dampens the affect of outliers.
(B) Linear models are robust to outliers.
(C) The parameters in logistic regression tend to take small values due to the nature of the problem setting and hence outliers get translated to the same range as other samples.
(D) The given statement is false.

Q12 Suppose I have 10,000 emails in my mailbox out of which 200 are spams. The spam detection system detects 150 mails as spams, out of which 50 are actually spams. What is the precision and recall of my spam detection system ?
(A) Precision = 33.333%, Recall = 25%
(B) Precision = 25%, Recall = 33.33%
(C) Precision = 33.33%, Recall = 75%
(D) Precision = 75%, Recall = 33.33%

Q13 Which of the following are true? TP True Positive, TN True Negative, FP False Positive, FN False Negative
(A) $\text{Precision} = \frac{\text{TP}}{\text{TP+FP}}$
(B) $\text{Recall} = \frac{\text{TP}}{\text{TP+FN}}$
(C) $\text{Accuracy} = \frac{2(\text{TP} + \text{TN})}{\text{TP+TN +FP+FN}}$
(D) $\text{Recall} = \frac{\text{FP}}{\text{TP+FP}}$

Q14 Suppose you are working on a binary classification problem using logistic regression. The logistic regression model is represented by the equation:
Logit (p) = $b_0 + b_1x_1 + b_2x_2 + \ldots\ldots b_nx_n$
Where,

- Logit(p) is the log-odds of the probability p belonging to the positive class.
- $x_1, x_2, \ldots\ldots x_n$ are the feature values.
- $b_0, b_1, b_2, \ldots\ldots b_n$ are the coefficients.

Given the following logistic model:
Logit(p) = $-2 + 1.5x_1 - 0.8x_2$
Find out the probability p when $x_1 = 3$ and $x_2 = 2$.
(A) 0.394 (B) 0.711
(C) 0.22 (D) 0.98

Q15 The following data is to be used to construct a regression model:

X	10	10	10	10	10	10
Y	2	4	9	9	3	3

The value of the slope is_____.
(A) –2.167 (B) –1.25
(C) 1.25 (D) None

Q16 Under what conditions does fitting a linear function in the leaf nodes of a regression tree instead of the constant function not increase the variance of the model?

(A) When the function is globally linear and each leaf node has at least 2 data points.
(B) When the function is locally linear and each leaf node has at least 2 data points.
(C) There is no change in model variance.
(D) None of the above.

Q17 In a classification dataset, there are 400 labeled examples belonging to 5 distinct classes. If each class has an equal number of examples, how many examples are there in each class?

(A) 80 (B) 100
(C) 200 (D) 400

Q18 In a logistic regression model, $b_1 = 2$, $b_0 = 1$, what is the estimated increase in the odds of the outcome for a one-unit increase in the predictor variable?

Q19 In a simple linear regression model, if the coefficient of determination (R2) is 0.64, what percentage of the variation in the dependent variable is explained by the independent variable (in %)?

Answer Key

Q1	0.5~0.5
Q2	C
Q3	C
Q4	0.587~0.587
Q5	A, C
Q6	C
Q7	A, B, D
Q8	B, D
Q9	C
Q10	C
Q11	A
Q12	A
Q13	A, B
Q14	B
Q15	D
Q16	A
Q17	A
Q18	2.5~2.5
Q19	64~64

Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

Y	F(X)	Loss
Cat	Cate	0
Cat	Dog	1
Dog	Panda	1
Panda	Panda	0
Rat	Dog	1
Rat	Rat	0

0-1 loss fxn

if error loss = 1

if error = 0 loss = 0

So, mean loss = $\frac{3}{6} = \frac{1}{2} = 0 \cdot 5$

Q2 Text Solution:

Tire Receiver Operating Characteristic (ROC) curve represents tire performance of a binary classification model across various classification thresholds. It is a graphical representation that shows the trade-off between tire True Positive Rate (Sensitivity) and the False Positive Rate as the threshold for classifying positive and negative instances is varied.

The ROC curve is a valuable tool for evaluating and comparing the performance of classification models, especially in situations where the balance between True Positives and False Positives is critical. It provides insights into how well the model can discriminate between tire two classes at different decision boundaries (thresholds).

Options (a), (b) and (d) do not accurately describe the ROC curve or its purpose, making them incorrect.

Q3 Text Solution:

$$\text{Precision} = \frac{\text{TP}}{\text{Total predicted P}}$$

$$= \frac{\text{TP}}{\text{TP+FP}} = \frac{120}{120+25} = 0.82$$

Q4 Text Solution:

$$\text{P} = \frac{\text{TP}}{\text{TP} + \text{FP}} = \frac{175}{175+123} = 0.587$$

Q5 Text Solution:

Type 1 error   False positive

Type 2 error   False Neagtive

Q6 Text Solution:

Fatigve : Positive

Not fatige : Negative

- FP : Driver is not fatigve but classifier say he is fative $\neq$0
- FN: Driver fatigve, but classfier say no FN = O

Recall = 1

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

we want FN = 0

Q7 Text Solution:

- Learning to drive using a reward signal = Supervised learning
- Predicting disease from blood sample. = Supervised learning
- Face recognition to unlock your phone. = Supervised learning

Q8 Text Solution:

Predict the disease of a patient

Classification problem

Has the student Learned to play a guitar yes/No Classification problem

Q9 Text Solution:

In machine learning, the **validation dataset** is used to fine-tune the model's hyperparameters and prevent overfitting. It helps in selecting the best model by evaluating different hyperparameter settings before testing on the unseen **test dataset**.

- **Training Dataset** Used to train the model (Option A is incorrect).

- **Validation Dataset** Used to tune hyperparameters (Correct answer: Option C).
- **Test Dataset** Used to evaluate model performance (Option B is incorrect)

Q10 Text Solution:

$$\begin{bmatrix} 1 & \\ 1 & \\ \vdots & x \\ 1 & \end{bmatrix}$$

design matrix

10000 × 18

Q11 Text Solution:

. The squashing of output values between [0, 1] dampens the effect of outliers. -While the sigmoid function squashes outputs, outliers still affect the decision boundary during training.

Q12 Text Solution:

200 spam Actual

150 dredic spam   50 True spam

100 False spam

$$\mathrm{P} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} = \frac{50}{50+100} = \frac{1}{3}$$

$$\mathrm{R} = \frac{\mathrm{TP}}{\text{Total actual P}} = \frac{50}{200} = \frac{1}{4}$$

Q13 Text Solution:

Precision = TP / (TP + FP)

- **Precision** measures the proportion of correctly predicted positive cases out of all predicted positives.
- Formula: **Precision = TP / (TP + FP)**

Recall = TP / (TP + FN)

- **Recall (Sensitivity)** measures how many actual positives were correctly predicted.
- Formula: **Recall = TP / (TP + FN)**

Q14 Text Solution:

$$\mathrm{P} = \frac{1}{1+e^{-xi\beta}}$$

$$= \frac{1}{1+e^{-(-2+1.5x_1 \ldots\ldots 8x_2)}} = 0.711$$

Q15 Text Solution:

$$m = \frac{\mathrm{cov}(x,y)}{\sigma_x^2}$$

Here, $\sigma_x^2 = 0$

Q16 Text Solution:

When the function is globally linear and each leaf node has at least 2 data points. =If the function is **globally linear**, using a **single linear model** (instead of a tree) would be better. This is **not a valid justification** for using local linear models in tree leaves.

Q17 Text Solution:

Number of data points in a clam = $\frac{400}{5} = 80$

Q18 Text Solution:

loge (odds) = $(x_i\beta)$

• loge (odds) = $\beta_0 + \beta_\iota x$

$$odd = e^{\beta_0+\beta_\iota x}$$

If x inc by 1 odds value =$e^{\beta_0+\beta_\iota(x+1)}$

$$\frac{odd\ new}{oddold} = \frac{C^{\beta_0+\beta_{,1}(x+1)}}{e^{\beta_0+\beta_1 x}} = e^{\beta_,}$$

$$\boxed{\frac{odd\ new}{oddold} = e^{\beta_1}}$$

$$\boxed{\text{odd new} = \mathrm{e}^2\text{odd old}}$$

Q19 Text Solution:

The coefficient of determination (R^2) represents the proportion of the variance in the dependent variable that is predictable from the independent variable. In this case, R^2 = 0.64 implies that 64% of the variation in the dependent variable is explained by the independent variable.